

Lab Eleven – Point Estimation and Resampling

Tips

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with “**Solution:**”.
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use “enter” to make your code readable, so it doesn’t run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you’re submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Don’t forget that you can copy-and-paste code when you’re doing something similar as we did in class!
- Make sure to plot all computed probabilities. This is good **ggplot** practice AND will help make sure you don’t make a mistake when computing the probability.

1 Introduction

When performing point estimation, there isn’t just one *right* answer. For example, the Method of Moments (MOM) and Maximum Likelihood Estimates (MLEs) aren’t always the same.

One question that comes up, is why we calculate the sample variance as

$$S_{n-1}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

when it is supposed to be the “average squared distance from the mean, i.e.,

$$S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

2 Comparing Estimators

2.1 Initial Simulation

In this lab, we will explore why we use S_{n-1}^2 and not S_n^2 . To do this, we will generate data from four known distributions:

- Poisson($\lambda = 2$)
- Binomial($n = 50, p = \frac{5-\sqrt{21}}{10}$)
- Normal($\mu = 0, \sigma = \sqrt{2}$)
- Exponential($\lambda = \frac{1}{\sqrt{2}}$).

Remark: Note that I have selected the parameters so that the true variance for all four models is 2.

For each distribution, write a loop that completes the following 1000 times. Put `set.seed(7272)` at the top of your code chunk.

- Generate a sample of size n from the distribution.
- Calculate and save S_{n-1}^2 and S_n^2 .

Start with $n = 20$. Is there evidence that S_{n-1}^2 is a better estimator than S_n^2 ?

Create a plot or combination of plots (using patchwork) to show the results across all four distributions. Make sure to create a corresponding table that numerically summarizes the results of the simulation. You should include

$$\text{Bias} = \text{Average of Estimates} - \text{True Value}$$

$$\text{Precision} = \frac{1}{\text{Variance of Estimates}}$$

$$\text{Mean Square Error} = \text{Bias}^2 + \text{Variance of Estimates}.$$

Solution:

Poisson:

```

1 set.seed(7272)
2
3 var.n1 = NA #initialized vector for S_{n-1}^2
4 var.n = NA #initialized vector for S_n^2
5 n = 20
6 var.true = 2
7
8 for(i in 1:1000){
9   x = rpois(n = n, lambda = 2)
10  mean.val = sum(x) / n
11  sqr.dev = (x-mean.val)^2
12
13  var.n1[i] = sum(sqr.dev) / (n-1)
14  var.n[i] = sum(sqr.dev) / n
15 }
16
17 # Compare estimators
18 bias.n1 = mean(var.n1 - var.true)
19 bias.n = mean(var.n - var.true)
20
21 precision.n1 = 1/var(var.n1)
22 precision.n = 1/var(var.n)
23
24 mse.n1 = bias.n1^2 + var(var.n1)
25 mse.n = bias.n^2 + var(var.n)
26
27 # Create numerical table
28 poisson.comparison = tibble(
29   Estimator = c("Variance (n-1)", "Variance (n)"),
30   Bias = c(bias.n1, bias.n),
31   Precision = c(precision.n1, precision.n),
32   MSE = c(mse.n1, mse.n)
33 )
34
35 poisson.comparison |>
36   knitr::kable(digits = 2, caption = "Poisson Variance Estimator Comparison")

```

Table 1: Poisson Variance Estimator Comparison

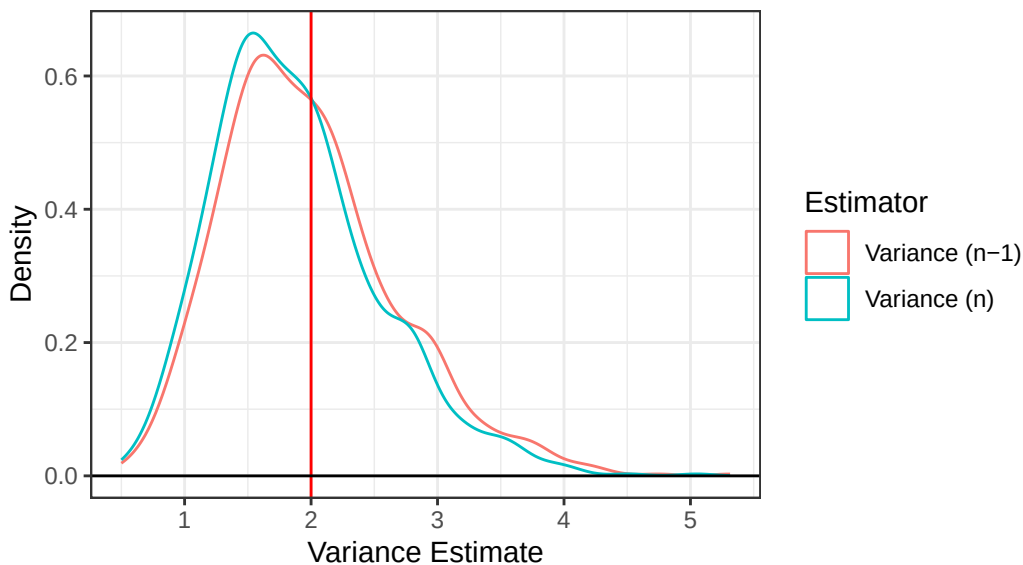
Estimator	Bias	Precision	MSE
Variance (n-1)	-0.02	2.10	0.48
Variance (n)	-0.12	2.33	0.44

```

1 # Create Plots
2 ggdat = tibble(
3   value = c(var.n1, var.n),
4   estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
5 )
6
7 ggdat |> ggplot() +
8   geom_density(aes(x=value, color=estimator))+
9   geom_vline(xintercept = var.true, col = "red") +
10  geom_hline(yintercept = 0) +
11  theme_bw() +
12  labs(x = "Variance Estimate",
13       y = "Density",
14       color = "Estimator") +
15  ggtitle("Distribution of Variance Estimators\n(Poisson, Lambda=2)")

```

Distribution of Variance Estimators
(Poisson, Lambda=2)



Binomial:

```

1 set.seed(7272)
2
3 var.n1 = NA #initialized vector for S_{n-1}^2
4 var.n = NA #initialized vector for S_n^2
5 n = 20
6 var.true = 2
7
8 for(i in 1:1000){
9   x = rbinom(n,50,(5-sqrt(21))/10)
10  mean.val = sum(x) / n
11  sqr.dev = (x-mean.val)^2
12
13  var.n1[i] = sum(sqr.dev) / (n-1)
14  var.n[i] = sum(sqr.dev) / n
15 }
16
17 # Compare estimators

```

```

18 bias.n1 = mean(var.n1 - var.true)
19 bias.n = mean(var.n - var.true)
20
21 precision.n1 = 1/var(var.n1)
22 precision.n = 1/var(var.n)
23
24 mse.n1 = bias.n1^2 + var(var.n1)
25 mse.n = bias.n^2 + var(var.n)
26
27 # Create numerical table
28 binom.comparison = tibble(
29   Estimator = c("Variance (n-1)", "Variance (n)"),
30   Bias = c(bias.n1, bias.n),
31   Precision = c(precision.n1, precision.n),
32   MSE = c(mse.n1, mse.n)
33 )
34
35 binom.comparison |>
36   knitr::kable(digits = 2, caption = "Binomial Variance Estimator Comparison")

```

Table 2: Binomial Variance Estimator Comparison

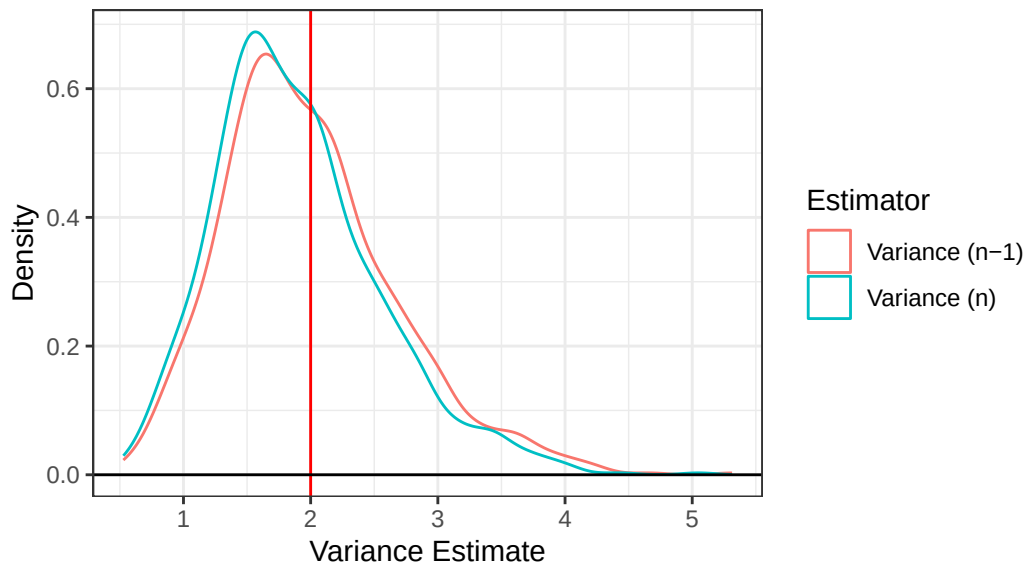
Estimator	Bias	Precision	MSE
Variance (n-1)	-0.01	2.13	0.47
Variance (n)	-0.11	2.36	0.44

```

1 # Create Plots
2 ggdat = tibble(
3   value = c(var.n1, var.n),
4   estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
5 )
6
7 ggdat |> ggplot() +
8   geom_density(aes(x=value, color=estimator))+
9   geom_vline(xintercept = var.true, col = "red") +
10  geom_hline(yintercept = 0) +
11  theme_bw() +
12  labs(x = "Variance Estimate",
13       y = "Density",
14       color = "Estimator") +
15  ggtitle("Distribution of Variance Estimators\n(Binomial, n=50, p=(5-sqrt(21))/10)")

```

Distribution of Variance Estimators (Binomial, $n=50$, $p=(5-\sqrt{21})/10$)



Normal:

```

1 set.seed(7272)
2
3 var.n1 = NA #initialized vector for S_{n-1}^2
4 var.n = NA #initialized vector for S_n^2
5 n = 20
6 var.true = 2
7
8 for(i in 1:1000){
9   x = rnorm(n,0,sqrt(2))
10  mean.val = sum(x) / n
11  sqr.dev = (x-mean.val)^2
12
13  var.n1[i] = sum(sqr.dev) / (n-1)
14  var.n[i] = sum(sqr.dev) / n
15 }
16
17 # Compare estimators
18 bias.n1 = mean(var.n1 - var.true)
19 bias.n = mean(var.n - var.true)
20
21 precision.n1 = 1/var(var.n1)
22 precision.n = 1/var(var.n)
23
24 mse.n1 = bias.n1^2 + var(var.n1)
25 mse.n = bias.n^2 + var(var.n)
26
27 # Create numerical table
28 norm.comparison = tibble(
29   Estimator = c("Variance (n-1)", "Variance (n)"),
30   Bias = c(bias.n1, bias.n),
31   Precision = c(precision.n1, precision.n),
32   MSE = c(mse.n1, mse.n)
33 )
34

```

```

35 norm.comparison |>
36 knitr::kable(digits = 2, caption = "Normal Dist. Variance Estimator Comparison")

```

Table 3: Normal Dist. Variance Estimator Comparison

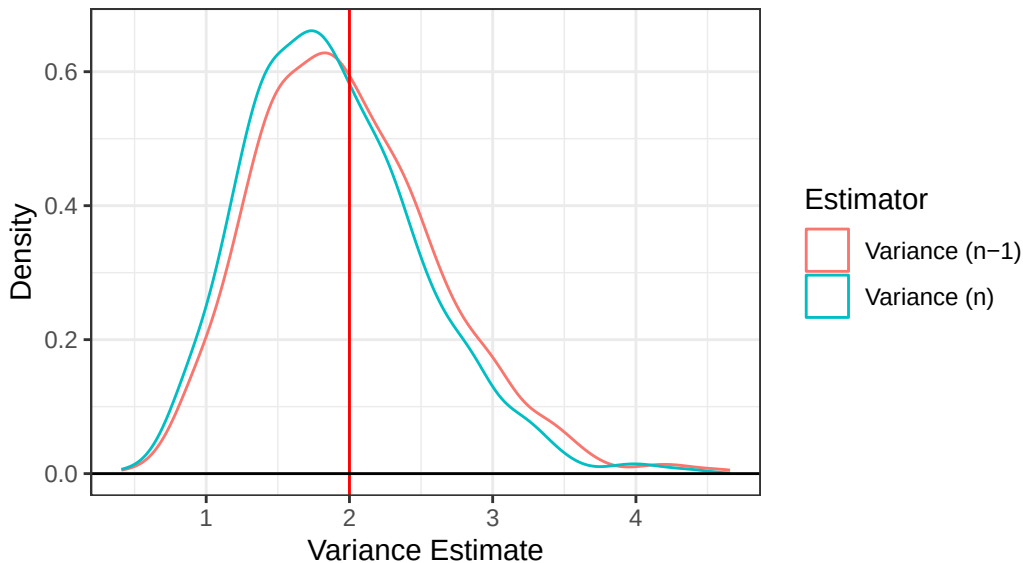
Estimator	Bias	Precision	MSE
Variance (n-1)	-0.01	2.41	0.41
Variance (n)	-0.11	2.67	0.39

```

1 # Create Plots
2 ggdat = tibble(
3   value = c(var.n1, var.n),
4   estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
5 )
6
7 ggdat |> ggplot() +
8   geom_density(aes(x=value, color=estimator))+
9   geom_vline(xintercept = var.true, col = "red") +
10  geom_hline(yintercept = 0) +
11  theme_bw() +
12  labs(x = "Variance Estimate",
13       y = "Density",
14       color = "Estimator") +
15  ggtitle("Distribution of Variance Estimators\n(Normal, mu=0, p=sqrt(2))")

```

Distribution of Variance Estimators
(Normal, $\mu=0$, $\sigma=\sqrt{2}$)



Exponential:

```

1 set.seed(7272)
2
3 var.n1 = NA #initialized vector for S_{n-1}^2
4 var.n = NA #initialized vector for S_n^2
5 n = 20
6 var.true = 2
7

```

```

8 for(i in 1:1000){
9   x = rexp(n,rate=1/sqrt(2))
10  mean.val = sum(x) / n
11  sqr.dev = (x-mean.val)^2
12
13  var.n1[i] = sum(sqr.dev) / (n-1)
14  var.n[i] = sum(sqr.dev) / n
15 }
16
17 # Compare estimators
18 bias.n1 = mean(var.n1 - var.true)
19 bias.n = mean(var.n - var.true)
20
21 precision.n1 = 1/var(var.n1)
22 precision.n = 1/var(var.n)
23
24 mse.n1 = bias.n1^2 + var(var.n1)
25 mse.n = bias.n^2 + var(var.n)
26
27 # Create numerical table
28 exp.comparison = tibble(
29   Estimator = c("Variance (n-1)", "Variance (n)"),
30   Bias = c(bias.n1, bias.n),
31   Precision = c(precision.n1, precision.n),
32   MSE = c(mse.n1, mse.n)
33 )
34
35 exp.comparison |>
36   knitr::kable(digits = 2, caption = "Exp. Distr. Variance Estimator Comparison")

```

Table 4: Exp. Distr. Variance Estimator Comparison

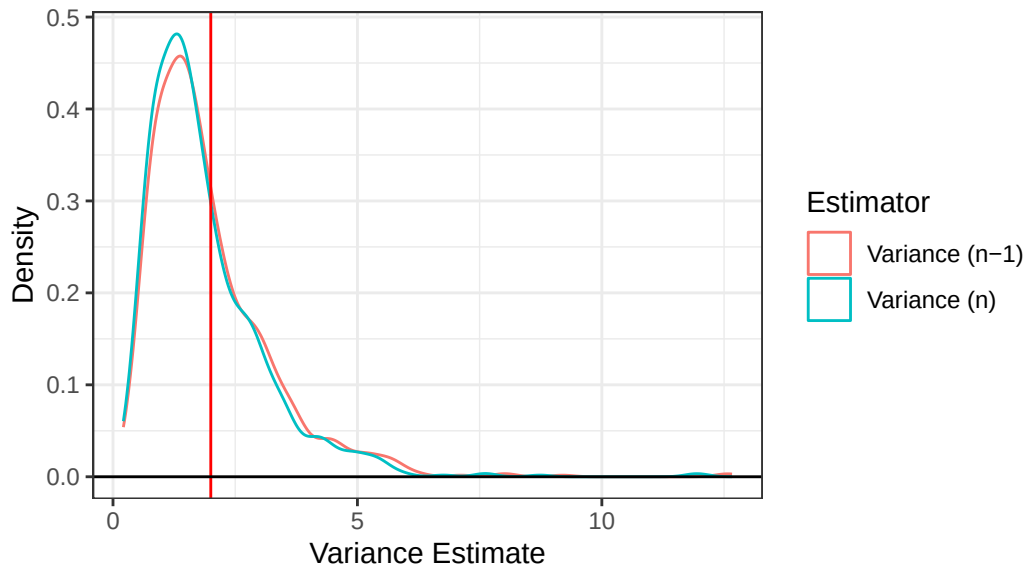
Estimator	Bias	Precision	MSE
Variance (n-1)	-0.03	0.61	1.64
Variance (n)	-0.12	0.68	1.49

```

1 # Create Plots
2 ggdat = tibble(
3   value = c(var.n1, var.n),
4   estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
5 )
6
7 ggdat |> ggplot() +
8   geom_density(aes(x=value, color=estimator))+
9   geom_vline(xintercept = var.true, col = "red") +
10  geom_hline(yintercept = 0) +
11  theme_bw() +
12  labs(x = "Variance Estimate",
13       y = "Density",
14       color = "Estimator") +
15  ggtitle("Distribution of Variance Estimators\n(Exponential, lambda = 1/sqrt(2))")

```

Distribution of Variance Estimators (Exponential, $\lambda = 1/\sqrt{2}$)



2.2 Simulation over various sample sizes

Consider the effect of the sample size have on these results. Provide a graphical and numerical summary of what happens as n increases – say $n = 20$, $n = 50$, $n = 100$, and $n = 500$. Does the result in your initial simulation hold across samples? Or, do things change as the sample size changes? Ensure to set your randomization seed.

Solution:

Note: I used `echo=FALSE` to make my PDF easier to look at whilst working on it (I took a “brute force” approach to this section). Apologies if this is an issue.

Generalizing the previous work into a function (code hidden):

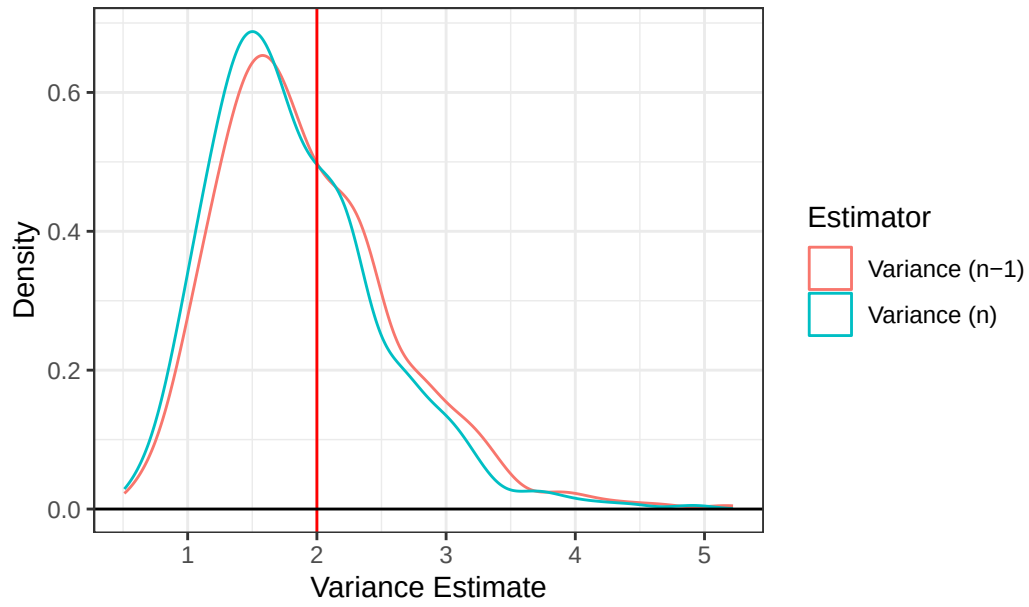
Running the Simulations (code hidden): Apologies for the looooong brute force coding below. This was just the most intuitive way for me to generate everything!

Poisson

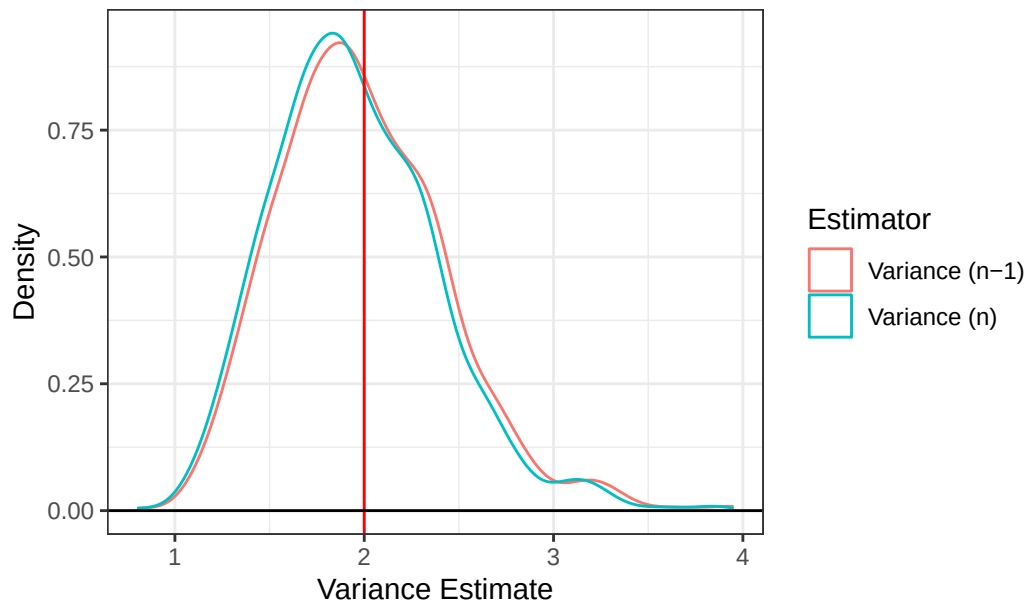
Table 5: Poisson Variance Estimators Across Sample Sizes

n	Bias (n-1)	Bias (n)	Precision (n-1)	Precision (n)	MSE (n-1)	MSE (n)
20	-0.0767	-0.1729	2.0849	2.3102	0.4855	0.4628
50	-0.0101	-0.0499	5.0170	5.2239	0.1994	0.1939
100	-0.0117	-0.0316	9.6444	9.8403	0.1038	0.1026
500	0.0050	0.0010	50.5581	50.7609	0.0198	0.0197

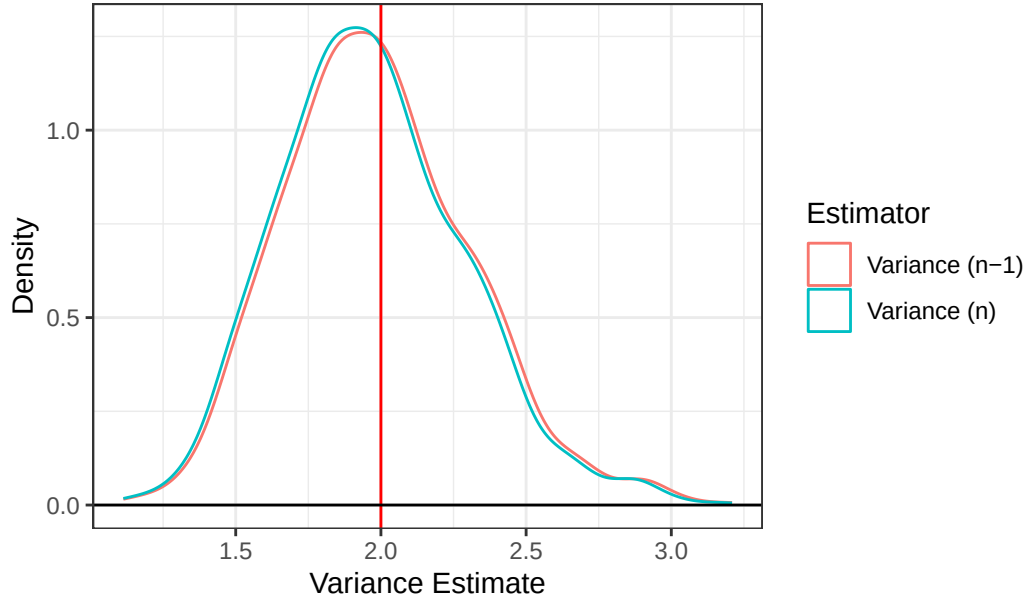
Poisson, $n = 20$



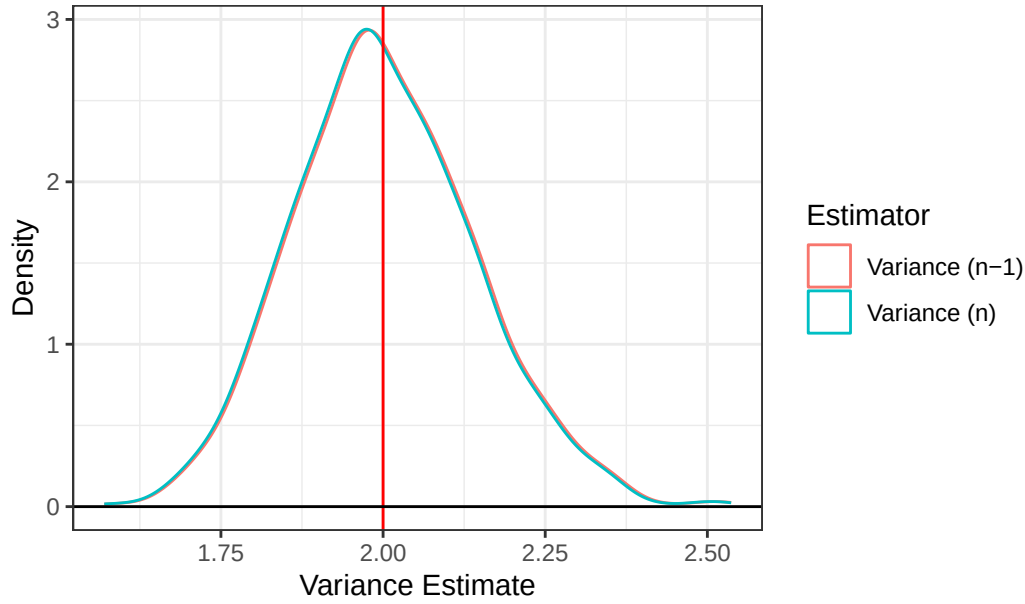
Poisson, $n = 50$



Poisson, $n = 100$



Poisson, $n = 500$

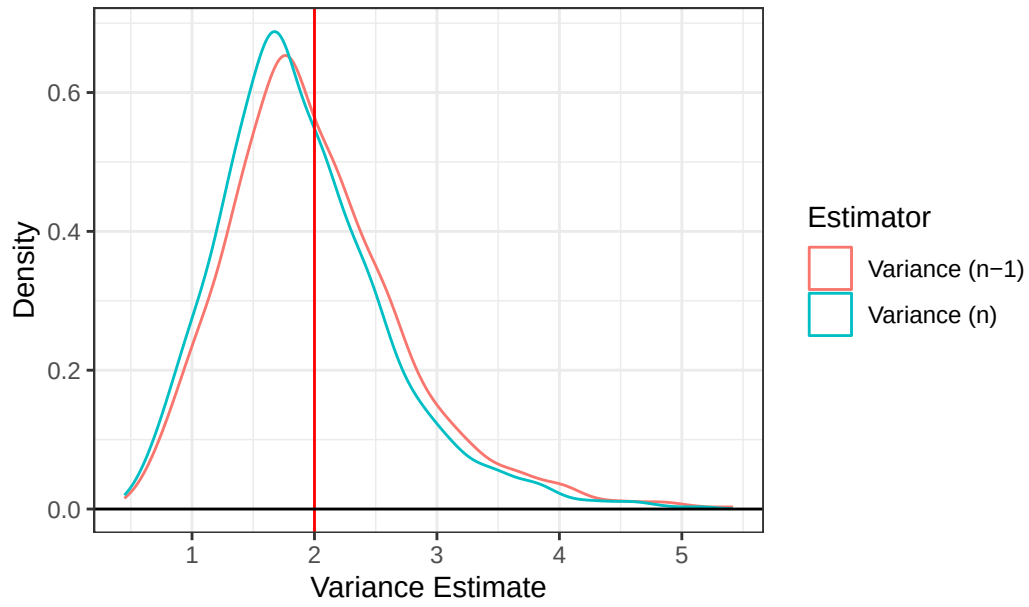


Binomial

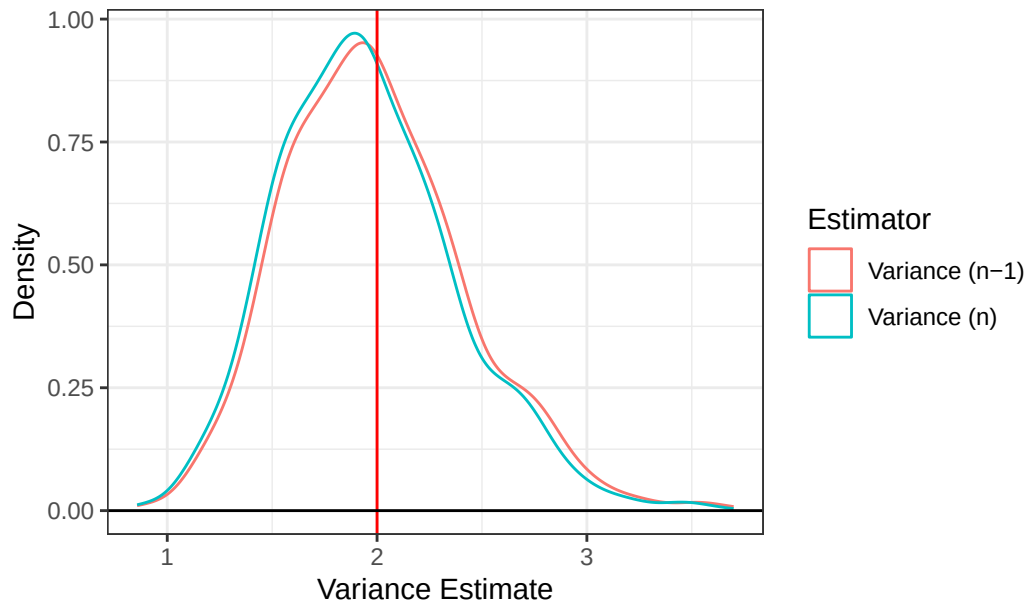
Table 6: Binomial Variance Estimators Across Sample Sizes

n	Bias ($n-1$)	Bias (n)	Precision ($n-1$)	Precision (n)	MSE ($n-1$)	MSE (n)
20	0.0043	-0.0959	1.8697	2.0717	0.5349	0.4919
50	-0.0075	-0.0473	5.3049	5.5236	0.1886	0.1833
100	-0.0107	-0.0306	9.6881	9.8848	0.1033	0.1021
500	0.0025	-0.0015	53.1977	53.4111	0.0188	0.0187

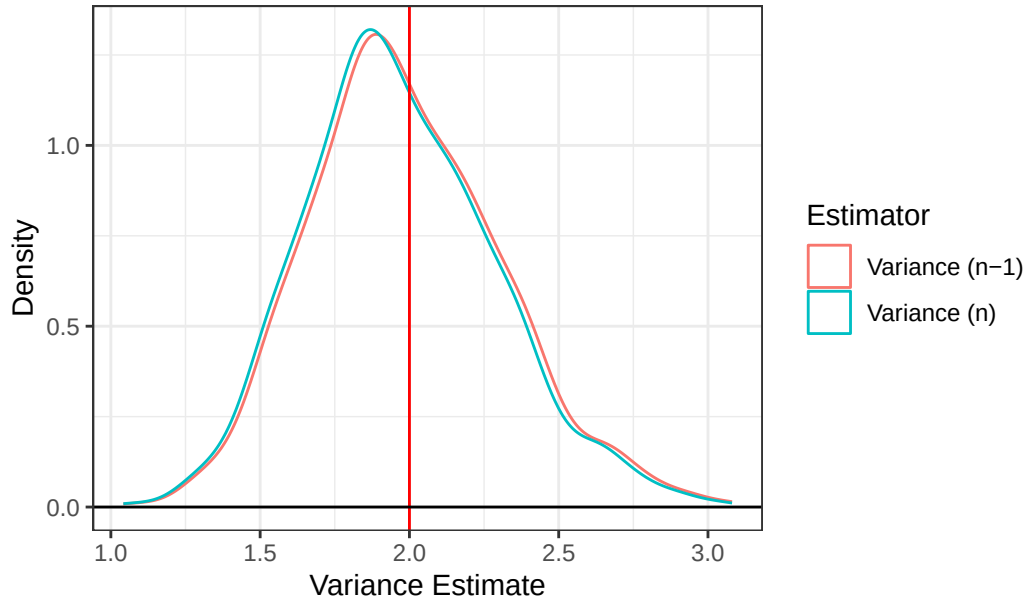
Binomial, $n = 20$



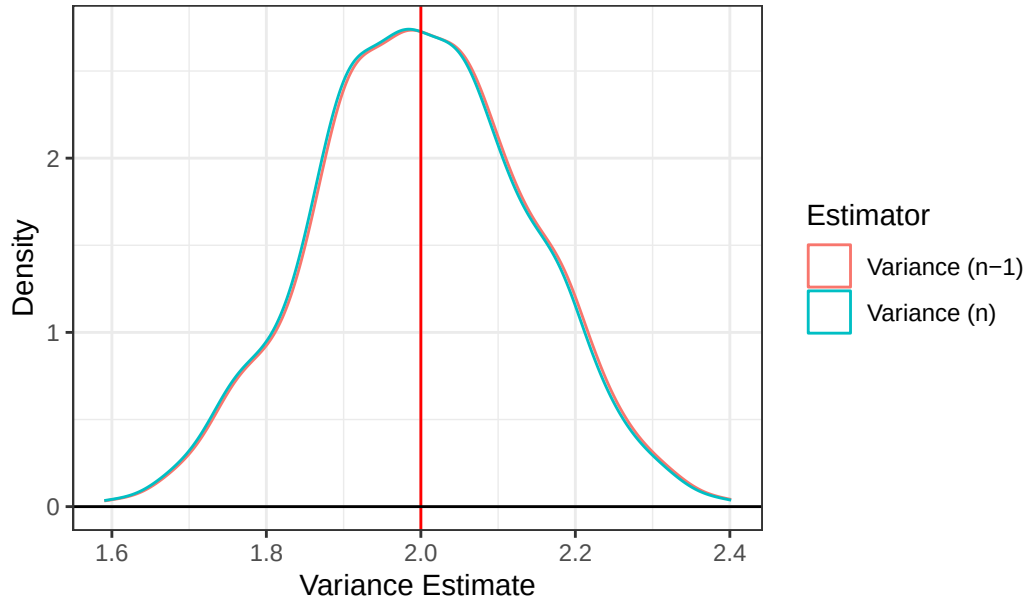
Binomial, $n = 50$



Binomial, $n = 100$



Binomial, $n = 500$

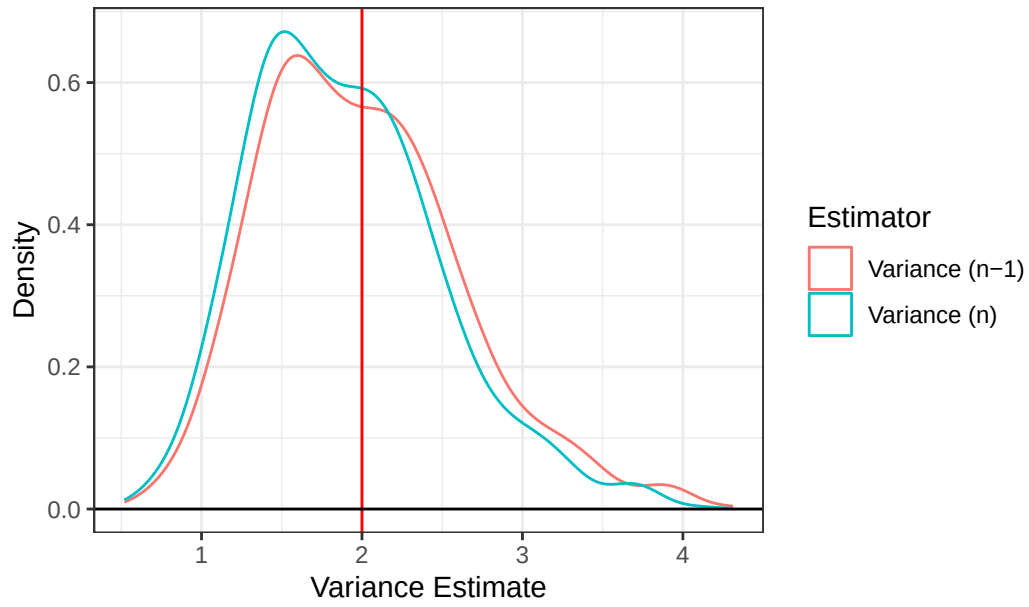


Normal

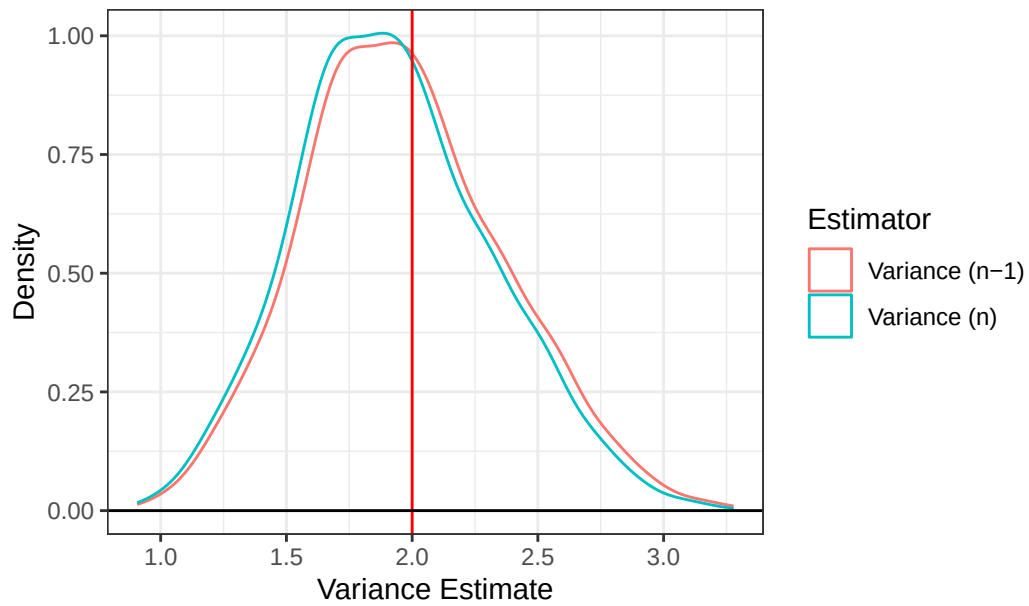
Table 7: Normal Distribution Variance Estimators Across Sample Sizes

n	Bias ($n-1$)	Bias (n)	Precision ($n-1$)	Precision (n)	MSE ($n-1$)	MSE (n)
20	0.0008	-0.0992	2.5488	2.8241	0.3923	0.3639
50	-0.0303	-0.0697	6.3219	6.5826	0.1591	0.1568
100	-0.0087	-0.0286	12.3458	12.5965	0.0811	0.0802
500	0.0051	0.0011	62.3261	62.5761	0.0161	0.0160

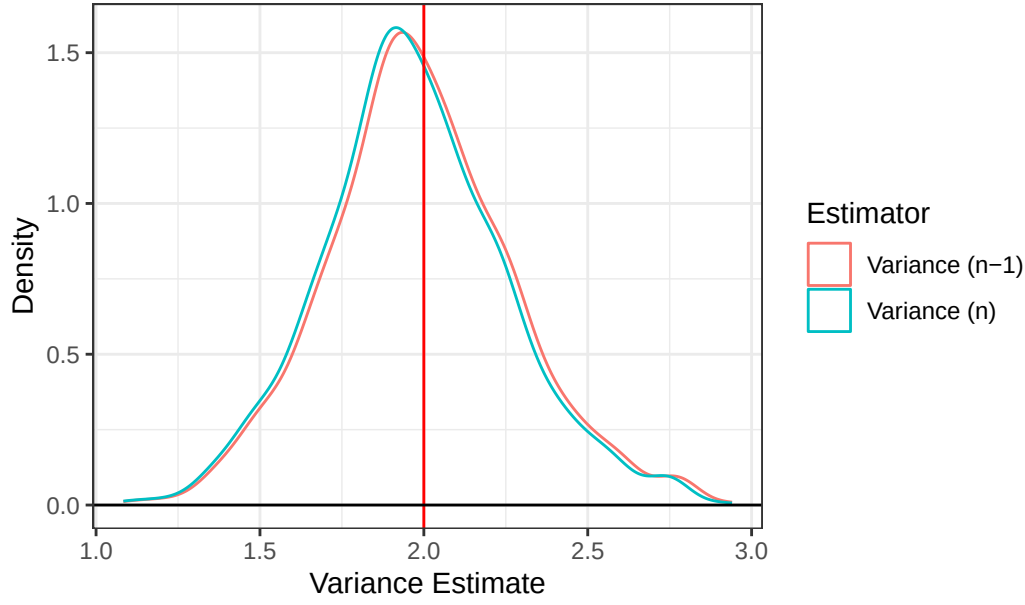
Normal, $n = 20$



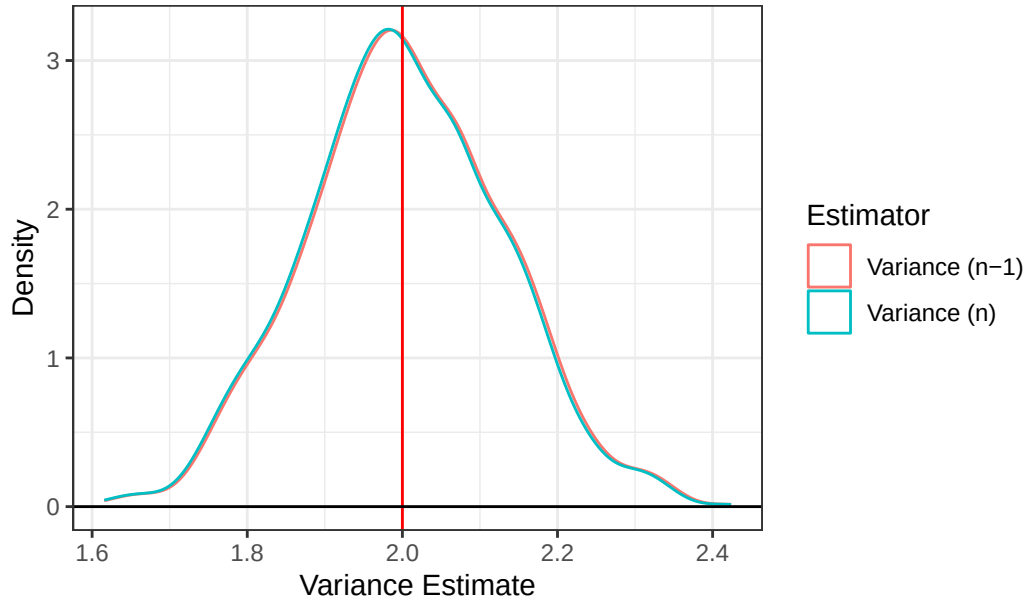
Normal, $n = 50$



Normal, $n = 100$



Normal, $n = 500$

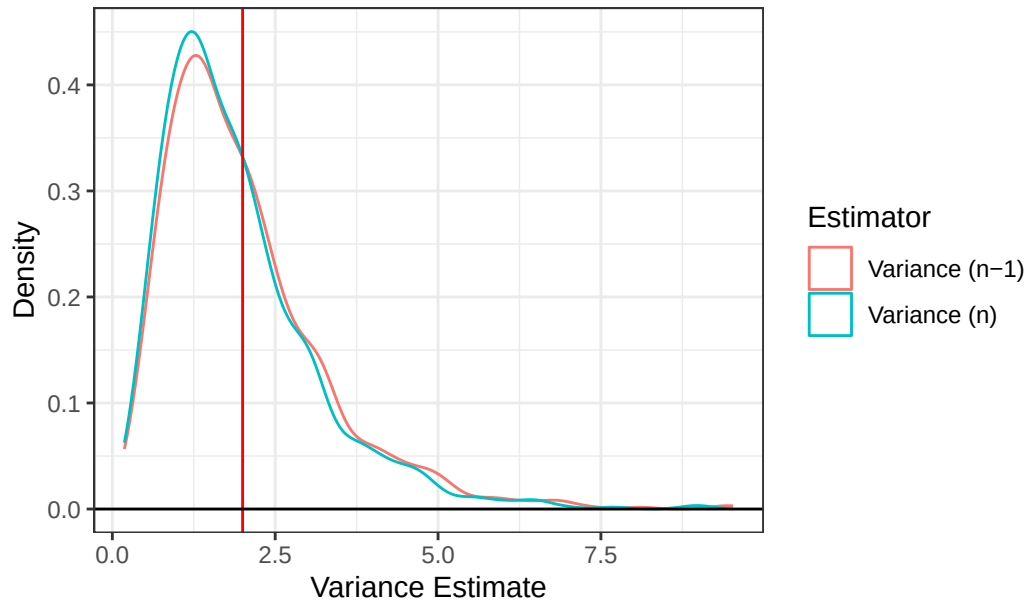


Exponential

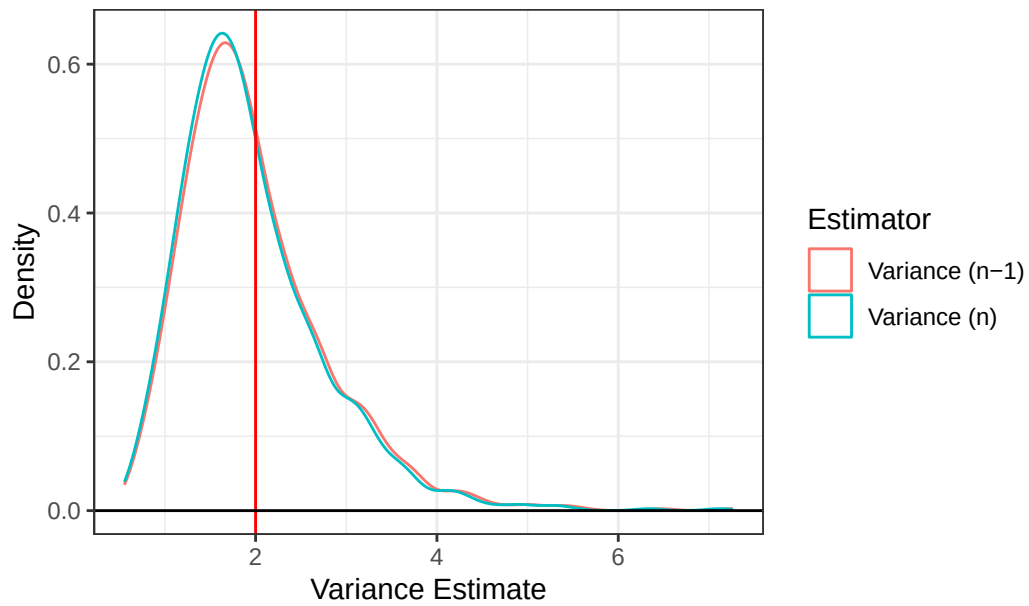
Table 8: Exponential Distribution Variance Estimators Across Sample Sizes

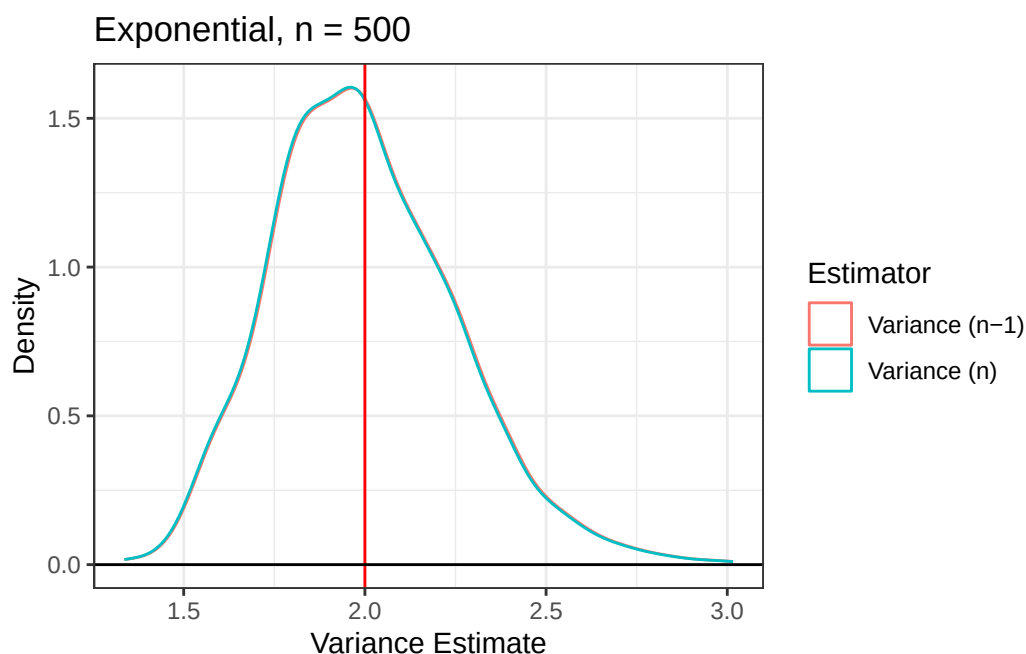
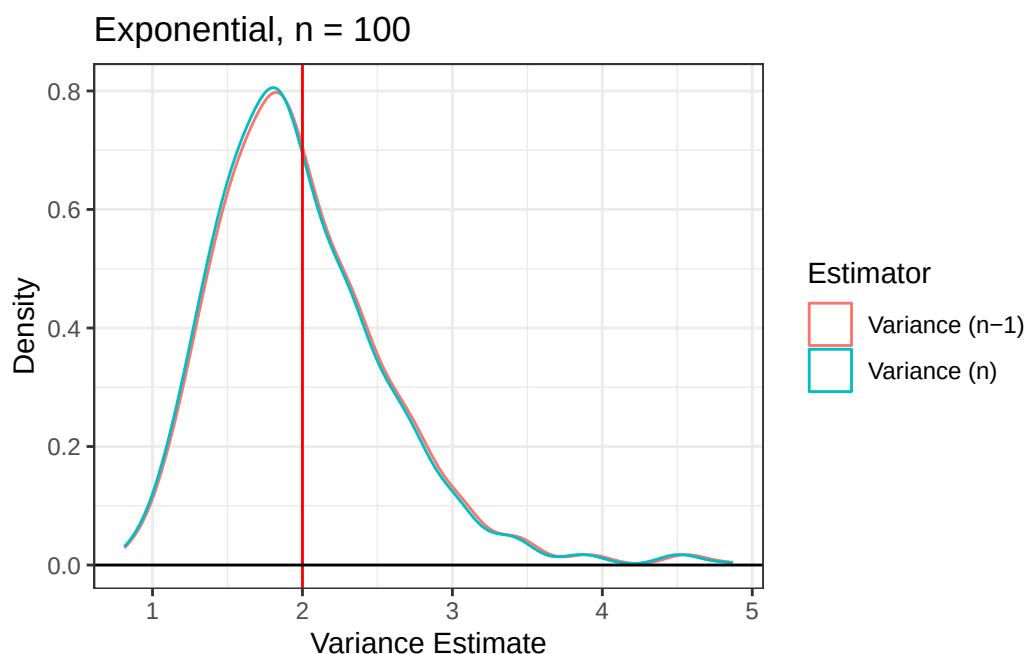
n	Bias ($n-1$)	Bias (n)	Precision ($n-1$)	Precision (n)	MSE ($n-1$)	MSE (n)
20	0.0185	-0.0824	0.6449	0.7145	1.5511	1.4063
50	-0.0066	-0.0465	1.5282	1.5912	0.6544	0.6306
100	-0.0060	-0.0260	2.9505	3.0104	0.3390	0.3329
500	0.0022	-0.0018	15.5934	15.6560	0.0641	0.0639

Exponential, $n = 20$



Exponential, $n = 50$





Commentary:

Across all four of the distributions, we find that the results change with sample size. Specifically, as n grows, the two density plots for each distribution (one for each variance measure) converge, and it becomes harder to differentiate between the density plot for s_{n-1}^2 and the density plot for s_n^2 . This phenomenon is further supported by the numerical summary tables. As n increases, we notice that bias and MSE for both s_{n-1}^2 and s_n^2 approach 0, while precision increases for both.

2.3 Simulation

You have written code to draw a random sample of various sizes from each distribution, and to calculate and store the calculated estimates. Rework your code to produce a density plot of the s_{n-1}^2 and s_n^2 . Do you have a guess about their distribution? Ensure to set your randomization seed.

NOTE: Plots are already generated above.

Commentary Density plots for the four distributions with $n = 100$, we can obtain a good estimate about the distributions for s_{n-1}^2 and s_n^2 . The first thing the density plots demonstrate is that s_{n-1}^2 and s_n^2 have very similar distributions (despite s_n^2 being slightly biased toward the left), though what distribution this is depends on the distribution of the model, and the sample size, n .

Small n: For small n , the distributions for s_{n-1}^2 and s_n^2 in all distributions are right-skewed.

Large n: With large enough n ($= 500$), s_{n-1}^2 and s_n^2 for the Poisson, Binomial, Normal, and Exponential distributions, all appear to approach a symmetric distribution. However, for the Exponential distribution, s_{n-1}^2 and s_n^2 still appears to follow a noticeably right-skewed distribution even at $n = 100$, taking a bit longer than the others to approach this symmetric shape.

The graphs also illustrate the reduced variance/improved precision as n increases, with observations further from the mean becoming less likely as n increases.

2.4 Bootstrap

Now, take *one* sample from each distribution. Instead of drawing a random sample at each iteration, sample *with replacement* from the one sample. Produce a density plot of the s_{n-1}^2 and s_n^2 . Do we get roughly the same information as we do in the full simulation? Try this over several seeds – what do you notice? Ensure to set your randomization seed.

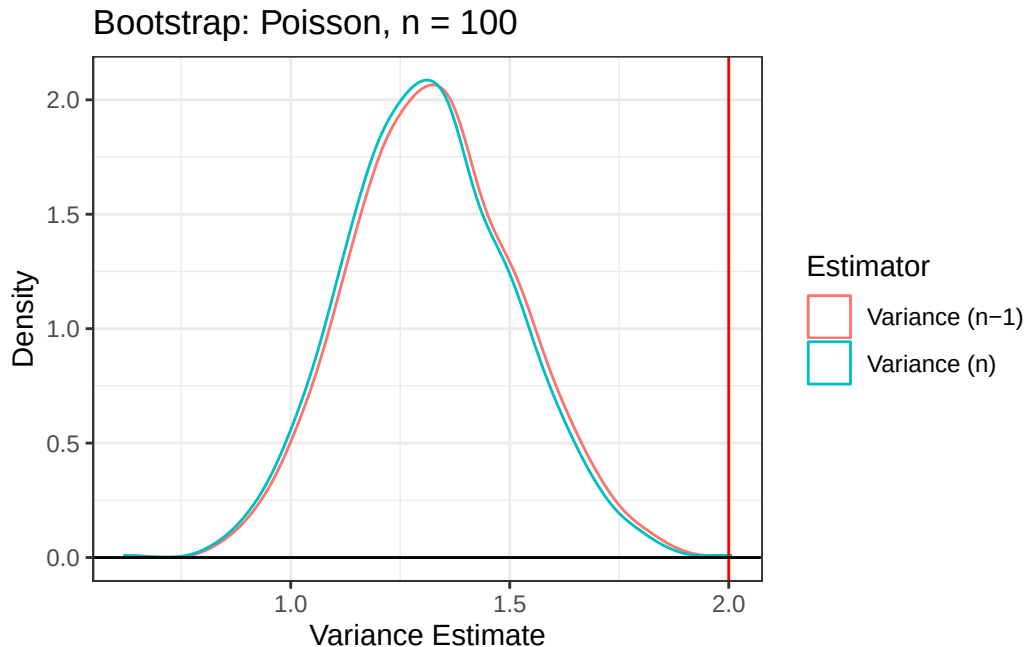
NOTE:

Turn off `#|echo=FALSE` to see my clunky code.

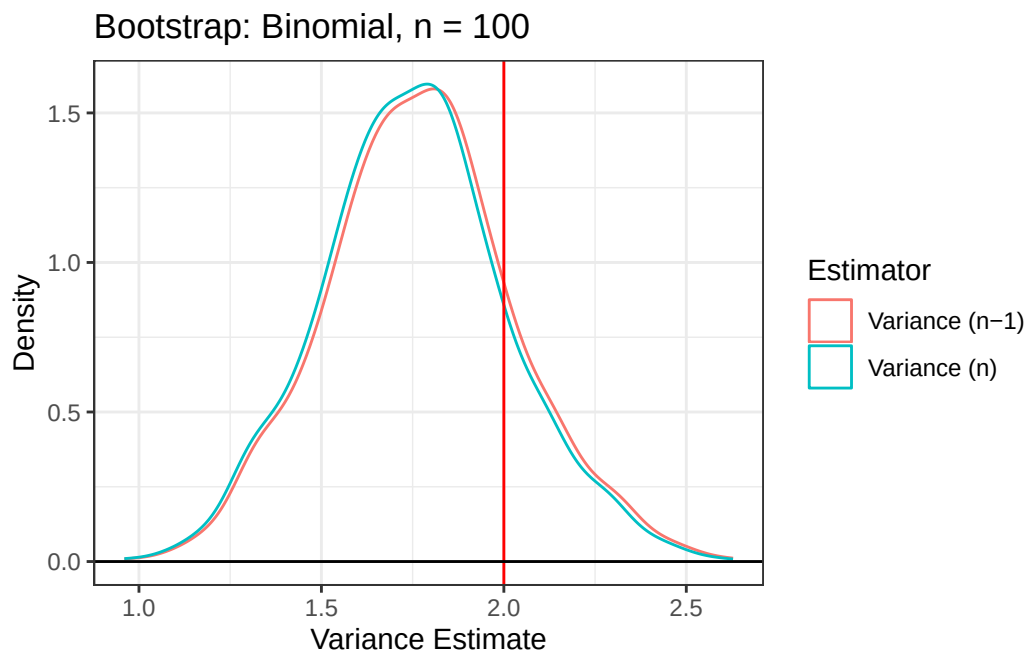
Table 9: Bootstrap Variance Estimators Across Distributions

Distribution	Bias (n-1)	Bias (n)	Precision (n-1)	Precision (n)	MSE (n-1)	MSE (n)
Poisson	-0.6716	-0.6849	27.3071	27.8616	0.4877	0.5050
Binomial	-0.2278	-0.2455	15.0863	15.3927	0.1182	0.1252
Normal	0.1063	0.0852	19.0837	19.4711	0.0637	0.0586
Exponential	-0.6416	-0.6552	17.4366	17.7906	0.4690	0.4855

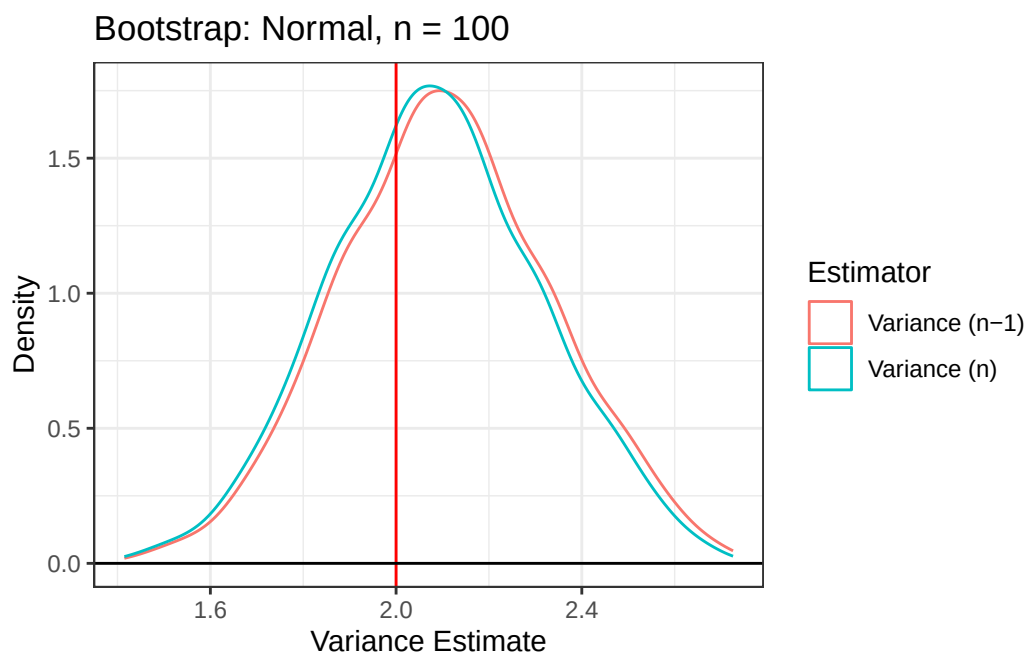
```
1 pois.boot$plot
```



```
1 binom.boot$plot
```

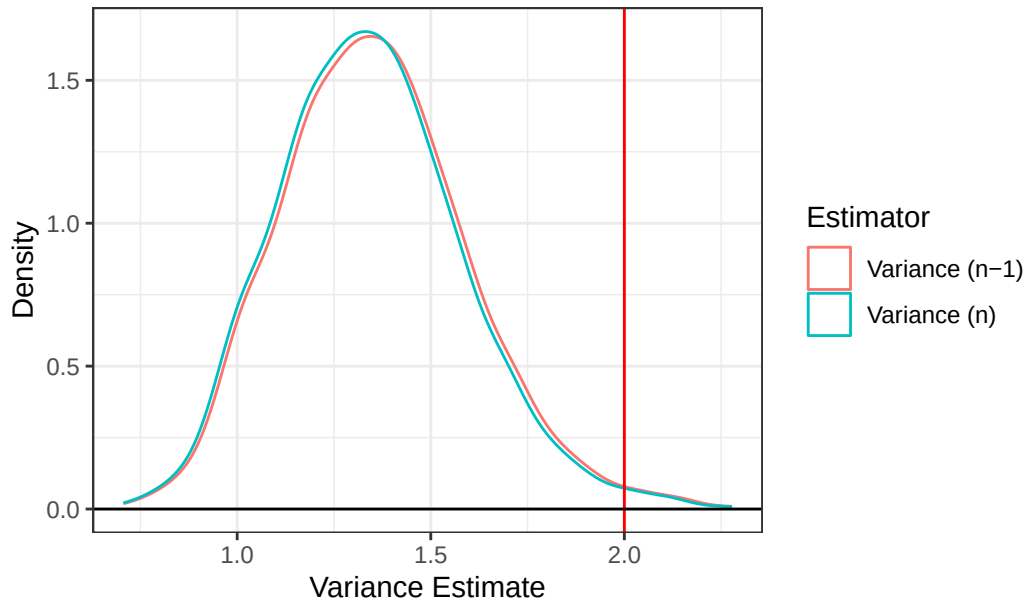


```
1 norm.boot$plot
```



```
1 exp.boot$plot
```

Bootstrap: Exponential, $n = 100$



3 Executive Summary

Summarize the work you've done here to provide a brief (200-300 word) summary of the simulation and the results using your work in Lab 11. Your summary should be professional, clear, and all claims should be corroborated with statistical support.